Behind the A: What NYC Restaurant Grades Tell You

Team name: Night's Watch

Members: Simone Wang (sw7317), Yabo Wang (yw10450), Yiyang Zhang (yz12500)

1. Problem Statement and Background

Every restaurant in New York has to post its letter grade by the door, and a lot of us use that card to decide where to eat. But the grade is a single letter, and it says very little about what the inspector actually found. In this project we want to look at what an A really tells a diner: how restaurants earn it, where and for which kinds of restaurants it is harder to get, which violations are behind low scores, and what happens to restaurants that fail.

We think this is worth doing because the data already shows some surprising things. Scores at first inspections bunch up right below the A cutoff. They peak at 12 and 13, the last two scores that still earn an A, and then fall from 4,784 inspections at 13 to just 215 at 14. About one in three routine inspections also finds mice, roaches or flies, or signs of them. Daniel Ho at Stanford has also questioned how consistent the city's inspection results are from one visit to the next. We are not trying to rank restaurants or blame inspectors. We just want readers to understand what the grade card in the window can and can't tell them.

Some context on how grading works (NYC Health, How We Score and Grade). Letter grades were introduced in 2010. Every violation adds points, so a lower score is better, and 0 to 13 points is an A. If a restaurant scores higher than 13 on its first inspection, it does not get a grade right away. An inspector comes back unannounced, usually within a month, and the second visit decides the grade (14 to 27 is a B and 28 or more is a C). Inspectors can also close a restaurant on the spot if they find a public health hazard. This means an A on the door could come from a clean first inspection, or from a failed first inspection followed by a better second one.

2. Dataset

We will use the DOHMH New York City Restaurant Inspection Results dataset (43nn-pn8j) from NYC Open Data. Inspectors from the NYC Department of Health and Mental Hygiene collect it during unannounced inspections. They record every violation they find, together with the score, the grade and any action taken, such as closing the restaurant. The city updates the file every day, and when we checked it on October 17, 2026 it had 296,819 rows covering 31,554 restaurants. Each row is one violation from one inspection. The file only includes restaurants that are currently open and roughly their last three years of inspections, and violations dismissed at a hearing are removed.

The attributes we plan to use are the restaurant ID (camis), name (dba) and cuisine (cuisine_description); location (boro, zipcode, latitude, longitude, and nta, a 2010 neighborhood code); the inspection itself (inspection_date, inspection_type, action); the result (score, grade, grade_date); and the violation (violation_code, violation_description, critical_flag).

Some preprocessing is needed before we can chart anything. We will save a dated copy of the CSV so our numbers don't shift while we work, and remove 3,908 restaurants that have not been inspected yet (they show the placeholder date 1900-01-01). Since the score is repeated on every violation row, we will split the data into restaurant, inspection and violation tables so each inspection is counted once. We will also group the 148 violation codes into five categories (pests, temperature, contamination, facilities and records), match each first inspection with the restaurant's re-inspection, and match each closure with its

reopening. For the chain question we need to merge spelling variants of the same brand, such as "DUNKIN" and "DUNKIN'". The map will use the 2010 neighborhood boundaries, since those match the nta codes in the data.

3. Domain Questions

Our questions start with what an A means, then look at where and for whom it is harder to get, then at why scores go up, and finally at what happens when a restaurant fails.

1. How does a restaurant earn its A, and how are first-inspection scores distributed? (a histogram of scores with the A cutoff marked)
2. How much does the share of A grades vary by borough and neighborhood? (a zoomable map)
3. How much do scores differ by cuisine? (median and spread of scores for the 20 most common cuisines)
4. Do branches of the same chain get similar grades? (one dot per branch for chains like Dunkin, Starbucks and McDonald's)
5. Which violations drive scores up, and what share of them are critical? (the most cited violations, split into critical and not critical)
6. Where do mice, roaches and flies show up most? (pest violations per inspection by neighborhood and by cuisine, to see whether place or food type matters more)
7. Do restaurants that miss an A on the first inspection recover at the re-inspection? (a flow from first score to final grade)
8. Which restaurants get closed, and how long does it take them to reopen? (closure reasons, days closed and failed reopening inspections)

Use of AI: we used Claude to help polish the writing, look for background sources and run some early checks on the data. The topic, the questions and the design decisions are our own, and we will double-check every number against our saved copy of the data.